# Tutorial 9


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Exercises 9.8 + 9.11 + 9.18

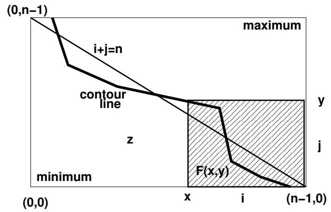


## ex. 9.8

The function $h : \mathbb{N}^2 \to \mathbb{Z}$ is ascending in both arguments. For a given constant $n \in \mathbb{N}$, specify and design a command to compute the number of points $(i,j) \in \mathbb{N}^2$ with $i + j < n$ and $h(i,j) > 0$.

We can specify this as follows:

**const** $n$ : $\mathbb{N}$;
**var** $z$ : $\mathbb{Z}$;
$\{P : \ Z = \#\{(i,j) \mid i,j : \ 0 \le i < n \ \wedge \ 0 \le j < n \ \wedge \ i+j < n \ \wedge \ h(i,j) > 0\}$
$T$;
$\{Q : \ Z = z\}$




Let $F(x,y) = \#\{(i,j) \mid i,j : \ x \le i \ \wedge \ 0 \le j < y \ \wedge \ i+j < n \ \wedge \ h(i,j) > 0\}$
So, we want to compute $F(0,n)$.



## ex. 9.8

First, we investigate what happens if we increment $x$.

$F(x,y)$
$=$ { definition $F$ }
$\#\{(i,j) \mid i,j : \ x \le i < n \ \wedge \ 0 \le j < y \ \wedge \ i+j < n \ \wedge \ h(i,j) > 0\}$
$=$ { **assume** $x < n$; split non-empty domain }
$\#\{(i,j) \mid i,j : \ x+1 \le i < n \ \wedge \ 0 \le j < y \ \wedge \ i+j < n \ \wedge \ h(i,j) > 0\} +$
$\#\{(x,j) \mid j : \ 0 \le j < y \ \wedge \ x+j < n \ \wedge \ h(x,j) > 0\}$
$=$ { definition $F$; calculus }
$F(x+1,y) + \#\{(x,j) \mid j : \ 0 \le j < y \ \wedge \ j < n-x \ \wedge \ h(x,j) > 0\}$
$=$ { • **case** $y \le n-x$ }
$F(x+1,y) + \#\{(x,j) \mid j : \ 0 \le j < y \ \wedge \ h(x,j) > 0\}$
$=$ { **assume** $y > 0$; $h(x,j)$ is ascending in $j$, so $h(x,y-1)$ is maximal;
**assume** $h(x,y-1) \le 0$, so $h(x,j) \le 0$ for all $0 \le j < y$ }
$= F(x+1,y)$

$=$ { • **case** $n-x < y$ }
$F(x+1,y) + \#\{(x,j) \mid j : \ 0 \le j < n-x \ \wedge \ h(x,j) > 0\}$
$=$ { $h(x,j)$ is ascending in $j$; Hence $n-x < y \Rightarrow h(x,n-x) \le h(x,y-1)$;
**assume** $h(x,y-1) \le 0$, so $h(x,j) \le 0$ for all $0 \le j < n-x < y$ }
$= F(x+1,y)$



## ex. 9.8

Next, we investigate what happens if we decrement $y$.

$F(x,y)$
$=$ { definition $F$ }
$\#\{(i,j) \mid i,j : \ x \le i < n \ \wedge \ 0 \le j < y \ \wedge \ i+j < n \ \wedge \ h(i,j) > 0\}$
$=$ { **assume** $y > 0$; split non-empty domain }
$\#\{(i,j) \mid i,j : \ x \le i < n \ \wedge \ 0 \le j < y-1 \ \wedge \ i+j < n \ \wedge \ h(i,j) > 0\} +$
$\#\{(i,y) \mid i : \ x \le i < n \ \wedge \ i+y-1 < n \ \wedge \ h(i,y-1) > 0\}$
$=$ { definition $F$; calculus; $y > 0 \Rightarrow n-y+1 \le n$ }
$F(x,y-1) + \#\{(i,y) \mid i : \ x \le i < n-y+1 \ \wedge \ h(i,y-1) > 0\}$
$=$ { • **case** $x < n-y+1$; $h(i,y-1)$ is ascending, so $h(x,y-1)$ is minimal;
**assume** $h(x,y-1) > 0$, so $h(i,y-1) > 0$ for all $x \le i$ }
$F(x,y-1) + \#\{(i,y) \mid i : \ x \le i < n-y+1\}$
$=$ { size half-open interval }
$F(x+1,y) + n - x - y + 1$

$=$ { • **case** $x \ge n-y+1$; empty domain }
$F(x,y-1)$

## ex. 9.8

So, $F(x,y)$ satisfies the recurrence

$$\begin{array}{rcl}
x \geq n \ \vee \ y \leq 0 & \Rightarrow & F(x,y) = 0 \\
x < n \wedge y > 0 \wedge h(x, y-1) \leq 0 & \Rightarrow & F(x,y) = F(x+1,y)x \\
x < n \wedge y > 0 \wedge x + y > n \wedge h(x, y-1) > 0 & \Rightarrow & F(x,y) = F(x, y-1) \\
x < n \wedge y > 0 \wedge x + y \leq n \wedge h(x, y-1) > 0 & \Rightarrow & F(x,y) = F(x, y-1) + n - x - y + 1
\end{array}$$



## ex. 9.8

$$F(x,y) \quad = \quad \#\{(i,j) \mid i,j : \ x \leq i < n \ \wedge \ 0 \leq j < y \ \wedge \ i + j < n \ \wedge \ h(i,j) > 0\}$$

We can reformulate the specification:

**const** $n : \ \mathbb{N}$;
**var** $z : \ \mathbb{Z}$;
$\{P : \ Z = F(0,n)\}$
$T$;
{Q: Z=z}

It is clear that $x \geq n \vee y \leq 0 \Rightarrow F(x,y) = 0$.

So, we choose the invariant and guard:

$$\begin{array}{ll}
J: & Z = z + F(x,y) \\
B: & x < n \ \wedge \ y > 0
\end{array}$$

$J \ \wedge \ \neg B$
$\equiv$ { definition $J$ and $B$; logic }
$Z = z + F(x,y) \ \wedge \ (x \geq n \ \vee \ y \leq 0)$
$\Rightarrow$ { base case recurrence; $F(x,y) = 0$ }
$Q : Z = z$



## ex. 9.8

2 Initialization of the invariant is easy:

$\{P : \ Z = F(0,n)\}$
(∗ calculus ∗)
$\{Z = 0 + F(0,n)\}$
$z := 0; \ x := 0; \ y := n;$
$\{J : \ Z = z + F(x,y)\}$

3 Variant function: We will increment $x$ and decrement $y$ as long as $x < n \ \wedge \ y > 0$, so we choose $\text{vf} = n - x + y \in \mathbb{Z}$. Since $B \Rightarrow n - x + y > 0$, it is clear that $J \ \wedge \ B \Rightarrow \text{vf} \geq 0$.



## ex. 9.8: Body: $\{J \ \wedge \ B \ \wedge \ \text{vf} = V\}$ $S$ $\{J \ \wedge \ \text{vf} < V\}$

$\{Z = z + F(x,y) \ \wedge \ x < n \ \wedge \ y > 0 \ \wedge \ n - x + y = V\}$
**if** $h(x, y-1) \leq 0$ **then**
$\{h(x,y-1) \leq 0 \ \wedge \ Z = z + F(x,y) \ \wedge \ x < n \ \wedge \ y > 0 \ \wedge \ n - x + y = V\}$
(∗ logic; recurrence for $F(x,y)$; prepare $x := x + 1$ ∗)
$\{Z = z + F(x+1,y) \ \wedge \ n - (x+1) + y < V\}$
$x := x + 1;$
$\{Z = z + F(x,y) \ \wedge \ n - x + y < V\}$
**else**
$\{h(x,y-1) > 0 \ \wedge \ Z = z + F(x,y) \ \wedge \ x < n \ \wedge \ y > 0 \ \wedge \ n - x + y = V\}$
**if** $x + y \leq n$ **then**
$\{h(x,y-1) > 0 \ \wedge \ x + y \leq n \ \wedge \ Z = z + F(x,y) \ \wedge \ x < n \ \wedge \ y > 0 \wedge n - x + y = V\}$
(∗ logic; recurrence for $F(x,y)$ ∗)
$\{Z = z + n - x - y + 1 + F(x,y-1) \ \wedge \ n - x + y = V\}$
$z := z + n - x - y + 1;$
$\{Z = z + F(x,y-1) \ \wedge \ n - x + y = V\}$
**else**
$\{h(x,y-1) > 0 \ \wedge \ x + y > n \ \wedge \ Z = z + F(x,y) \ \wedge \ x < n \ \wedge \ y > 0 \wedge n - x + y = V\}$
(∗ logic; recurrence for $F(x,y)$ ∗)
$\{Z = z + F(x,y-1) \ \wedge \ n - x + y = V\}$
**end** (∗ collect branches ∗)
$\{Z = z + F(x,y-1) \ \wedge \ n - x + y = V\}$
(∗ calculus; prepare $y := y - 1$ ∗)
$\{Z = z + F(x,y-1) \ \wedge \ n - x + y - 1 < V\}$
$y := y - 1;$
$\{Z = z + F(x,y) \ \wedge \ n - x + y < V\}$
**end** (∗ collect branches; definitions $J$ and vf ∗)
$\{J \ \wedge \ \text{vf} < V\}$

## ex. 9.8

5 Conclusion:

**const** $n : \mathbb{N}$;
**var** $x, y, z : \mathbb{Z}$;
  $\{P : Z = \#\{(i,j) \mid i,j : 0 \le i < n \wedge 0 \le j < n \wedge i+j < n \wedge h(i,j) > 0\}\}$
$z := 0$;
$x := 0$;
$y := n$;
  $\{J : Z = z + \#\{(i,j) \mid i,j : x \le i < n \wedge 0 \le j < y \wedge i+j < n \wedge h(i,j) > 0\}\}$
    $(*\ \mathrm{vf} : n - x + y\ *)$
**while** $x < n \wedge y > 0$ **do**
  **if** $h(x, y-1) \le 0$ **then**
    $x := x + 1$;
  **else**
    **if** $x + y \le n$ **then**
      $z := z + n - x - y + 1$;
    **end**;
    $y := y - 1$;
  **end**;
**end**;
  $\{Q : z = Z\}$



## ex. 9.11

Design a command to compute (note, minor modification):

$$\begin{array}{ll} & \#\{(i,j) \mid i,j : 0 < i \wedge 0 \le j \wedge i^2 + j^2 < w^2\} \\ = & \#\{(i,j) \mid i,j : 0 < i \le w \wedge 0 \le j < w \wedge i^2 + j^2 < w^2\} \end{array}$$

Note, we chose deliberately half-open intervals for $i$ and $j$.

Clearly, the expression $i^2 + j^2$ is increasing in $i$ and $j$.

Let $F(x,y) = \#\{(i,j) \mid i,j : x < i \le w \wedge 0 \le j < y \wedge i^2 + j^2 < w^2\}$

The problem is specified formally as follows:

**const** $w : \mathbb{N}$;
**var** $z : \mathbb{Z}$;
  $\{P : Z = F(0, w)\}$
$T$;
  $\{Q : Z = z\}$



## ex. 9.11

$$F(x,y) = \#\{(i,j) \mid i,j : x < i \le w \wedge 0 \le j < y \wedge i^2 + j^2 < w^2\}$$

It is clear that (empty domain): $x \ge w \vee y \le 0 \Rightarrow F(x,y) = 0$.
We investigate what happens if we increment $x$.

$$\begin{array}{ll} & F(x,y) \\ = & \{\ \text{definition } F\ \} \\ & \#\{(i,j) \mid i,j : x < i \le w \wedge 0 \le j < y \wedge i^2 + j^2 < w^2\} \\ = & \{\ \textbf{assume } x < w\text{; split non-empty domain}\ \} \\ & \#\{(i,j) \mid i,j : x+1 < i \le w \wedge 0 \le j < y \wedge i^2 + j^2 < w^2\} + \\ & \#\{j \mid j : 0 \le j < y \wedge (x+1)^2 + j^2 < w^2\} \\ = & \{\ \text{definition } F\ \} \\ & F(x+1, y) + \#\{j \mid j : 0 \le j < y \wedge (x+1)^2 + j^2 < w^2\} \\ = & \{\ \textbf{assume } y > 0\text{; } (x+1)^2 + j^2 \text{ is increasing in } j, \\ & \quad \text{so } (x+1)^2 + (y-1)^2 \text{ is the maximum;} \\ & \quad \textbf{assume} \text{ that this maximum is less than } w^2\} \\ & F(x+1, y) + \#\{j \mid j : 0 \le j < y\} \\ = & \{\ \text{size half-open interval}\ \} \\ & F(x+1, y) + y \end{array}$$



## ex. 9.11

$$F(x,y) = \#\{(i,j) \mid i,j : x < i \le w \wedge 0 \le j < y \wedge i^2 + j^2 < w^2\}$$

Nex, we investigate what happens if we decrement $y$.

$$\begin{array}{ll} & F(x,y) \\ = & \{\ \text{definition } F\ \} \\ & \#\{(i,j) \mid i,j : x < i \le w \wedge 0 \le j < y \wedge i^2 + j^2 < w^2\} \\ = & \{\ \textbf{assume } y > 0\text{; split non-empty domain}\ \} \\ & \#\{(i,j) \mid i,j : x < i \le w \wedge 0 \le j < y-1 \wedge i^2 + j^2 < w^2\} + \\ & \#\{i \mid i : x < 1 \le w \wedge i^2 + (y-1)^2 < w^2\} \\ = & \{\ \text{definition } F\ \} \\ & F(x, y-1) + \#\{i \mid i : x < i \le w \wedge i^2 + (y-1)^2 < w^2\} \\ = & \{\ \textbf{assume } x < w\text{; } i^2 + (y-1)^2 \text{ is increasing in } i, \\ & \quad \text{so } (x+1)^2 + (y-1)^2 \text{ is the minimum;} \\ & \quad \textbf{assume} \text{ that this minimum is greater than or equal to } w^2\} \\ & F(x, y-1) \end{array}$$

1 Choice invariant and guard:

$$J: \quad Z = z + F(x,y)$$
$$B: \quad x < w \ \wedge \ y > 0$$

$J \ \wedge \ \neg B$
$\equiv$ { definition $J$ and $B$; logic }
$Z = z + F(x,y) \ \wedge \ (x \geq w \ \vee \ y \leq 0)$
$\Rightarrow$ { base case recurrence; $F(x,y) = 0$ }
$Q : Z = z$

2 Initialization of the invariant is easy:

$\{P: \ Z = F(0,w)\}$
(∗ calculus ∗)
$\{Z = 0 + F(0,w)\}$
$z := 0; x := 0; \ y := w;$
$\{J: \ Z = z + F(x,y)\}$

3 Variant function: vf $= w - x + y \in \mathbb{Z}$. Clearly, $J \ \wedge \ B \Rightarrow$ vf $\geq 0$





4 Body of the loop: $\{J \ \wedge \ B \ \wedge \ \text{vf} = V\} \ S \ \{J \ \wedge \ \text{vf} < V\}$

$\{Z = z + F(x,y) \ \wedge \ x < w \ \wedge \ y > 0 \ \wedge \ w - x + y = V\}$
**if** $(x+1)^2 + (y-1)^2 < w^2$ **then**
$\{(x+1)^2 + (y-1)^2 < w^2 \ \wedge \ Z = z + F(x,y) \ \wedge \ x < w \ \wedge \ y > 0 \ \wedge \ w - x + y = V\}$
(∗ logic; recurrence for $F(x,y)$; prepare $x := x + 1$ ∗)
$\{Z = z + y + F(x+1,y) \ \wedge \ w - (x+1) + y < V\}$
$z := z + y;$
$\{Z = z + F(x+1,y) \ \wedge \ w - (x+1) + y < V\}$
$x := x + 1;$
$\{Z = z + F(x,y) \ \wedge \ w - x + y < V\}$
**else**
$\{(x+1)^2 + (y-1)^2 \geq w^2 \ \wedge \ Z = z + F(x,y) \ \wedge \ x < w \ \wedge \ y > 0 \ \wedge \ w - x + y = V\}$
(∗ logic; recurrence for $F(x,y)$; prepare $y := y - 1$ ∗)
$\{Z = z + F(x,y-1) \ \wedge \ w - x + y - 1 < V\}$
$y := y - 1;$
$\{Z = z + F(x,y) \ \wedge \ w - x + y < V\}$
**end** (∗ collect branches; definitions $J$ and vf ∗)
$\{J \ \wedge \ \text{vf} < V\}$





5 Conclusion:

**const** $w : \ \mathbb{Z};$
**var** $x, \ y, \ z : \ \mathbb{Z};$
$\{P: \ Z = \#\{(i,j) \mid i,j: \ 0 < i \ \wedge \ 0 \leq j \ \wedge \ i^2 + j^2 < w^2\} \ \}$
$z := 0;$
$x := 0;$
$y := w;$
$\{J: \ Z = z + \#\{(i,j) \mid i,j: \ x < i \leq w \ \wedge \ 0 \leq j < y \ \wedge \ i^2 + j^2 < w^2\} \ \}$
(∗ vf : $w - x + y$ ∗)
**while** $x < w \ \wedge \ y > 0$ **do** (∗ Note $(x+1)^2 + (y-1)^2 = x^2 + y^2 + 2(x - y + 1)$ ∗)
**if** $x * x + y * y + 2 * (x - y + 1) < w * w$ **then**
$z := z + y;$
$x := x + 1;$
**else**
$y := y - 1;$
**end**;
**end**;
$\{Q: \ z = Z\}$

Using this program fragment, it is easy to compute an approximation of $\pi$. The program traverses the boundary of a quarter of a circle, so if we choose a sufficiently large radius $w$, then the program approximates the area of a quarter of a circle with radius $w$, i.e. $\pi w^2/4$. So, we can approximate $\pi$ with the assignment $pi := 4 * z/(w * w)$ after finalization of the loop, where $pi \in \mathbb{R}$.





The function $f : \mathbb{N}^2 \to \mathbb{Z}$ is ascending in both arguments. For a given constants $n$ and $w$, specify and design a command to compute the number of points $(i,j) \in \mathbb{N}^2$ with $2 \cdot j \leq i < n$ and $f(i,j) < w$.

We can specify this as follows:

**const** $n : \ \mathbb{N}; \ w : \ \mathbb{Z};$
**var** $z : \ \mathbb{Z};$
$\{P: \ Z = \#\{(i,j) \mid i,j: \ 0 \leq j \ \wedge \ 2 \cdot j \leq i < n \ \wedge \ f(i,j) < w\}$
$T;$
$\{Q: \ Z = z\}$

Let $F(x,y) = \#\{(i,j) \mid i,j: \ y \leq j \ \wedge \ 2 \cdot j \leq i < x \ \wedge \ f(i,j) < w\}$
So, we can reformulate the specification as:

**const** $n : \ \mathbb{N}; \ w : \ \mathbb{Z};$
**var** $z : \ \mathbb{Z};$
$\{P: \ Z = F(n,0)\}$
$T;$
$\{Q: \ Z = z\}$

Clearly (due to an empty domain), $F(x,y) = 0 \Leftarrow x \leq 2 \cdot y$.
We choose $J: \ Z = z + F(x,y)$ and $B: \ x > 2 \cdot y$, such that $J \ \wedge \ \neg B \Rightarrow Q$.

$$F(x,y) = \#\{(i,j) \mid i,j:\ y \le j \ \wedge\ 2\cdot j \le i < x \ \wedge\ f(i,j) < w\}$$

We clearly want to decrement $x$ or increment $y$.
First, we investigate what happens if we decrement $x$.

$$\begin{array}{ll}
 & F(x,y) \\
= & \{\ \text{definition } F\ \} \\
 & \#\{(i,j) \mid i,j:\ y \le j \ \wedge\ 2\cdot j \le i < x \ \wedge\ f(i,j) < w\} \\
= & \{\ \textbf{assume } x > 2\cdot y \text{ (guard of the loop); split non-empty domain }\} \\
 & \#\{(i,j) \mid i,j:\ y \le j \ \wedge\ 2\cdot j \le i < x-1 \ \wedge\ f(i,j) < w\}\ + \\
 & \#\{j \mid j:\ y \le j \ \wedge\ 2\cdot j \le x-1 \ \wedge\ f(x-1,j) < w\} \\
= & \{\ \text{definition } F\ \} \\
 & F(x-1,y) + \#\{j \mid j:\ y \le j \ \wedge\ 2\cdot j \le x-1 \ \wedge\ f(x-1,j) < w\} \\
= & \{\ f(x-1,j) \text{ is ascending in } j \text{, so } f(x-1,y) \text{ is the minimum;} \\
 & \quad \textbf{assume } f(x-1,y) \ge w \text{, then } f(x-1,y) \ge w \text{ for all } j \ge y\} \\
 & F(x-1,y) + 0 \\
= & \\
 & F(x-1,y)
\end{array}$$





$$F(x,y) = \#\{(i,j) \mid i,j:\ y \le j \ \wedge\ 2\cdot j \le i < x \ \wedge\ f(i,j) < w\}$$

Next, we investigate what happens if we increment $y$.

$$\begin{array}{ll}
 & F(x,y) \\
= & \{\ \text{definition } F\ \} \\
 & \#\{(i,j) \mid i,j:\ y \le j \ \wedge\ 2\cdot j \le i < x \ \wedge\ f(i,j) < w\} \\
= & \{\ \textbf{assume } x > 2\cdot y \text{ (guard of the loop); split non-empty domain }\} \\
 & \#\{(i,j) \mid i,j:\ y+1 \le j \ \wedge\ 2\cdot j \le i < x \ \wedge\ f(i,j) < w\}\ + \\
 & \#\{i \mid i:\ 2\cdot y \le i < x \ \wedge\ f(i,y) < w\} \\
= & \{\ \text{definition } F\ \} \\
 & F(x,y+1) + \#\{i \mid i:\ 2\cdot y \le i < x \ \wedge\ f(i,y) < w\} \\
= & \{\ f(i,y) \text{ is ascending in } i \text{, so } f(x-1,y) \text{ is the maximum;} \\
 & \quad \textbf{assume } f(x-1,y) < w \text{, then } f(x-1,y) < w \text{ for all } i < x\} \\
 & F(x-1,y) + \#\{i \mid i:\ 2\cdot y \le i < x\} \\
= & \{\ \text{size of half-open interval }\} \\
 & F(x-1,y) + x - 2\cdot y
\end{array}$$





1 Choice invariant and guard:

$$\begin{array}{ll}
J: & Z = z + F(x,y) \\
B: & x > 2\cdot y
\end{array}$$

$$\begin{array}{ll}
 & J \ \wedge\ \neg B \\
\equiv & \{\ \text{definition } J \text{ and } B\text{; logic }\} \\
 & Z = z + F(x,y) \ \wedge\ \neg(x > 2\cdot y) \\
\Rightarrow & \{\ \text{base case recurrence; } F(x,y) = 0\ \} \\
 & Q : Z = z
\end{array}$$

2 Initialization of the invariant is easy:

$\{P:\ Z = F(n,0)\}$
$(*$ calculus $*)$
$\{Z = 0 + F(n,0)\}$
$z := 0; x := n;\ y := 0;$
$\{J:\ Z = z + F(x,y)\}$

3 Variant function: $\mathrm{vf} = x - 2\cdot y \in \mathbb{Z}$. Clearly, $B \Rightarrow J \ \wedge\ B \Rightarrow \mathrm{vf} \ge 0$





4 Body of the loop: $\{J \ \wedge\ B \ \wedge\ \mathrm{vf} = V\}\ S\ \{J \ \wedge\ \mathrm{vf} < V\}$

$\{Z = z + F(x,y) \ \wedge\ x > 2\cdot y \ \wedge\ x - 2\cdot y = V\}$
**if** $f(x-1,y) \ge w$ **then**
$\{f(x-1,y) \ge w \ \wedge\ Z = z + F(x,y) \ \wedge\ x > 2\cdot y \ \wedge\ x - 2\cdot y = V\}$
$(*$ logic; recurrence for $F(x,y)$; prepare $x := x-1$ $*)$
$\{Z = z + F(x-1,y) \ \wedge\ x - 1 - 2\cdot y < V\}$
$x := x - 1;$
$\{Z = z + F(x,y) \ \wedge\ x - 2\cdot y < V\}$
**else**
$\{f(x-1,y) < w \ \wedge\ Z = z + F(x,y) \ \wedge\ x > 2\cdot y \ \wedge\ x - 2\cdot y = V\}$
$(*$ logic; recurrence for $F(x,y)$; prepare $y := y+1$ $*)$
$\{Z = z + x - 2\cdot y + F(x,y+1) \ \wedge\ x - 2\cdot(y+1) < V\}$
$z := z + x - 2 * y;$
$\{Z = z + F(x,y+1) \ \wedge\ x - 2\cdot(y+1) < V\}$
$y := y + 1;$
$\{Z = z + F(x,y) \ \wedge\ x - 2\cdot y < V\}$
**end** $(*$ collect branches; definitions $J$ and vf $*)$
$\{J \ \wedge\ \mathrm{vf} < V\}$

5 Conclusion:

**const** $n: \mathbb{N}$; $w: \mathbb{Z}$;
**var** $x, y, z: \mathbb{Z}$;
  $\{P: Z = \#\{(i,j) \mid i,j: 0 \leq j \wedge 2 \cdot j \leq i < n \wedge f(i,j) < w\}$
$z := 0$;
$x := n$;
$y := 0$;
  $\{J: Z = z + \#\{(i,j) \mid i,j: y \leq j \wedge 2 \cdot j \leq i < x \wedge f(i,j) < w\}$
    $(*\ \text{vf}: x - 2 \cdot y\ *)$
**while** $x > 2 * y$ **do**
  **if** $f(x-1, y) \geq w$ **then**
    $x := x - 1$;
  **else**
  $z := z + x - 2 * y$;
  $y := y - 1$;
  **end**;
**end**;
  $\{Q: z = Z\}$